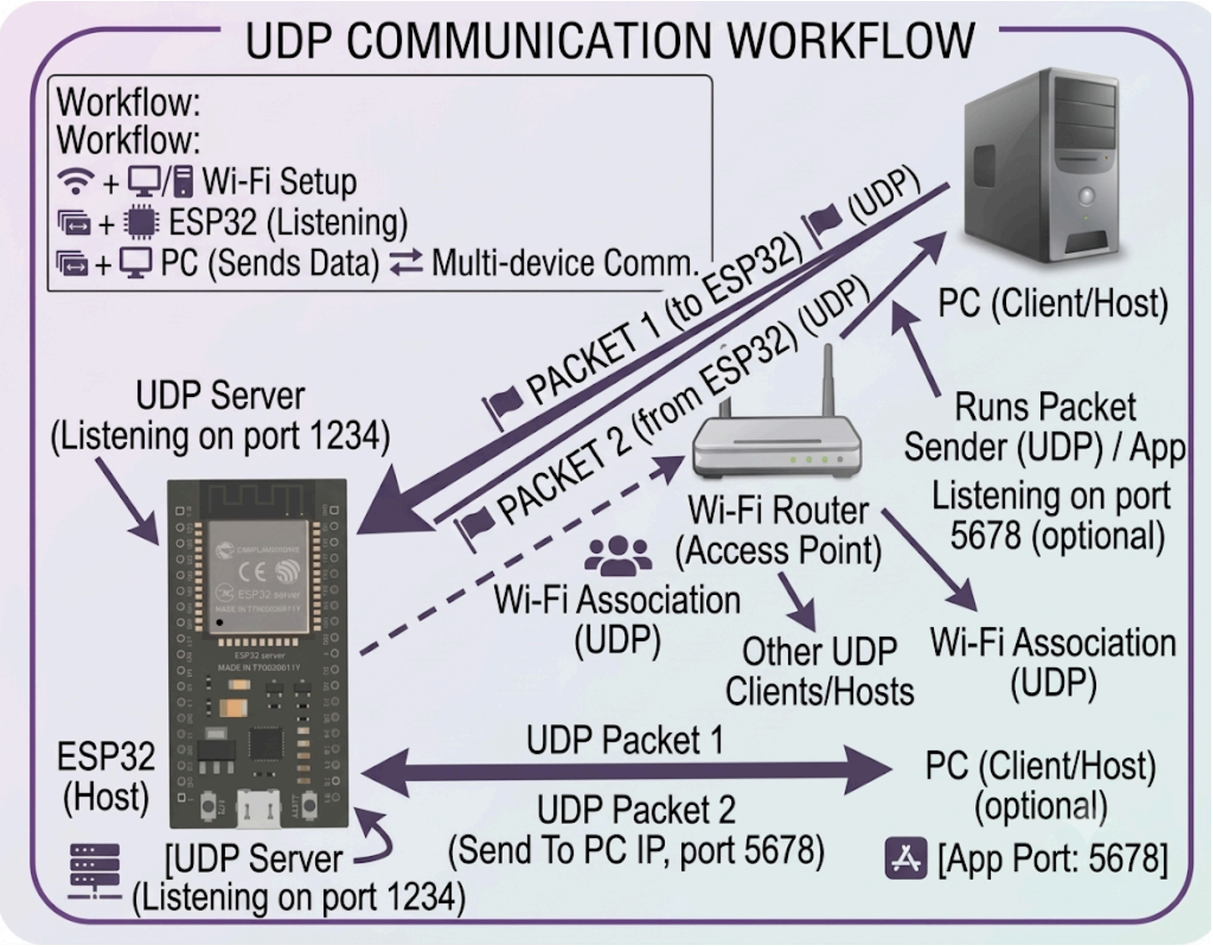
UDP Communication Protocol

Wireless Connectionless Communication over standard Wi-Fi Networks.

What Is UDP?

Wireless Connectionless Communication over standard Wi-Fi Networks.



Rides the Wi-Fi Radio

Uses standard Wi-Fi infrastructure (TCP/IP stack over 802.11) and IP network packets.

No Access Point Needed

Wi-Fi AP/Router Required (both devices must join the same Wi-Fi network)

Built for Low Latency

Designed for fast, lightweight datagram communication without connection handshakes.

Flexible Topology

IP-Based Architecture (supports Unicast and Broadcast via IP addresses and Ports).

Where Is UDP Used in IoT?

Network-integrated application cases

PC-to-ESP32 Telemetry

Local Web Application Interfaces

Smart Home Hub Integration

Real-Time Data Streaming over LAN.

Prerequisites & Hardware :

Hardware: One ESP32 board and a Computer/PC (connected to the same Wi-Fi network).

USB data cable: Required for programming and Serial Monitor access. Make sure the cable supports data, not only charging.

Power: USB power is sufficient for normal testing. A separate regulated supply can be used if the board's power-input requirements are satisfied.

Software: Keep Arduino IDE, but add Packet Sender (or Windows Terminal/winget) for testing UDP packets from your PC.

Network Requirements: 2.4 GHz Local Wi-Fi Network with valid SSID & Password.

Step 1: Install Packet Sender via Windows Terminal (winget install packetsender).

Step 2: Configure Arduino IDE and include <WiFi.h> and <WiFiUdp.h> libraries.

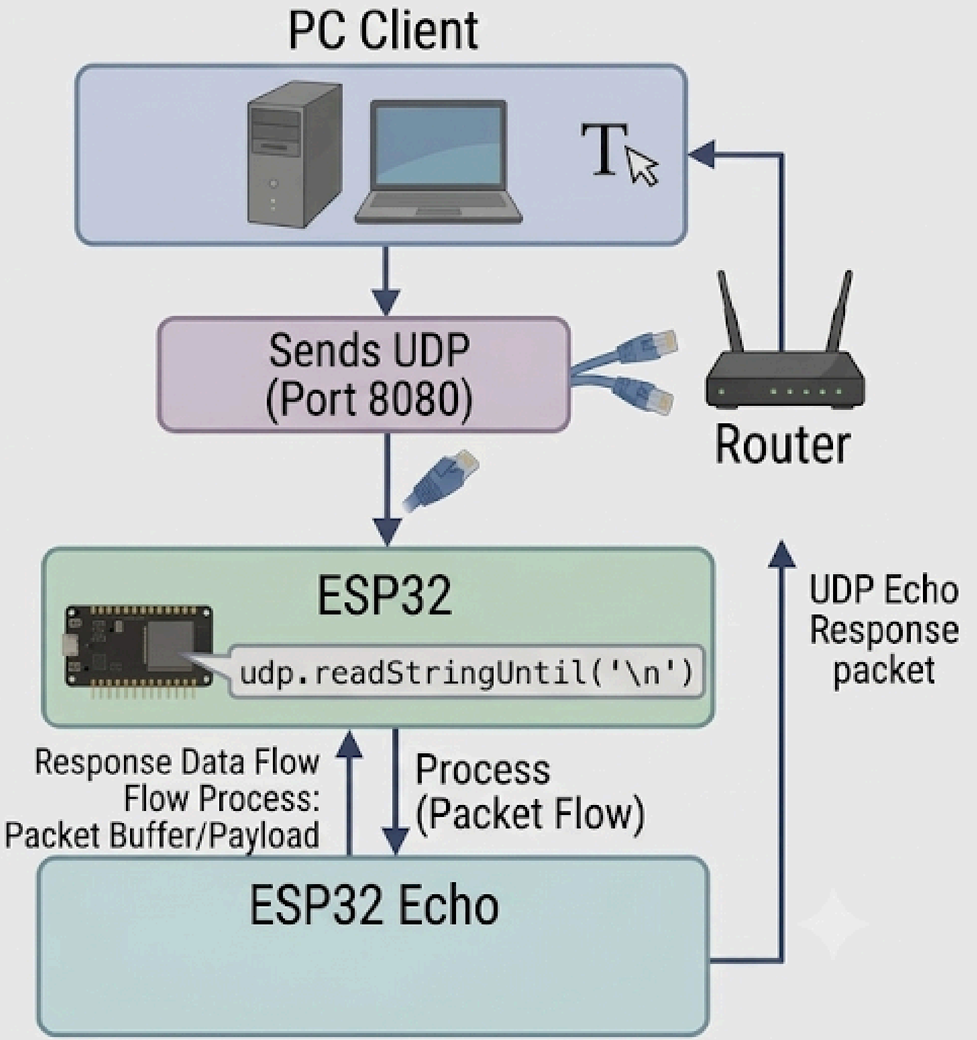
Step 3: Write and Upload UDP Echo Code (configure Wi-Fi credentials SSID, PASSWORD, and target localPort).

Step 4: Obtain ESP32 IP Address via Serial Monitor.

Step 5: Configure Packet Sender (Set IP address, UDP port 4210, select UDP protocol, and send message)

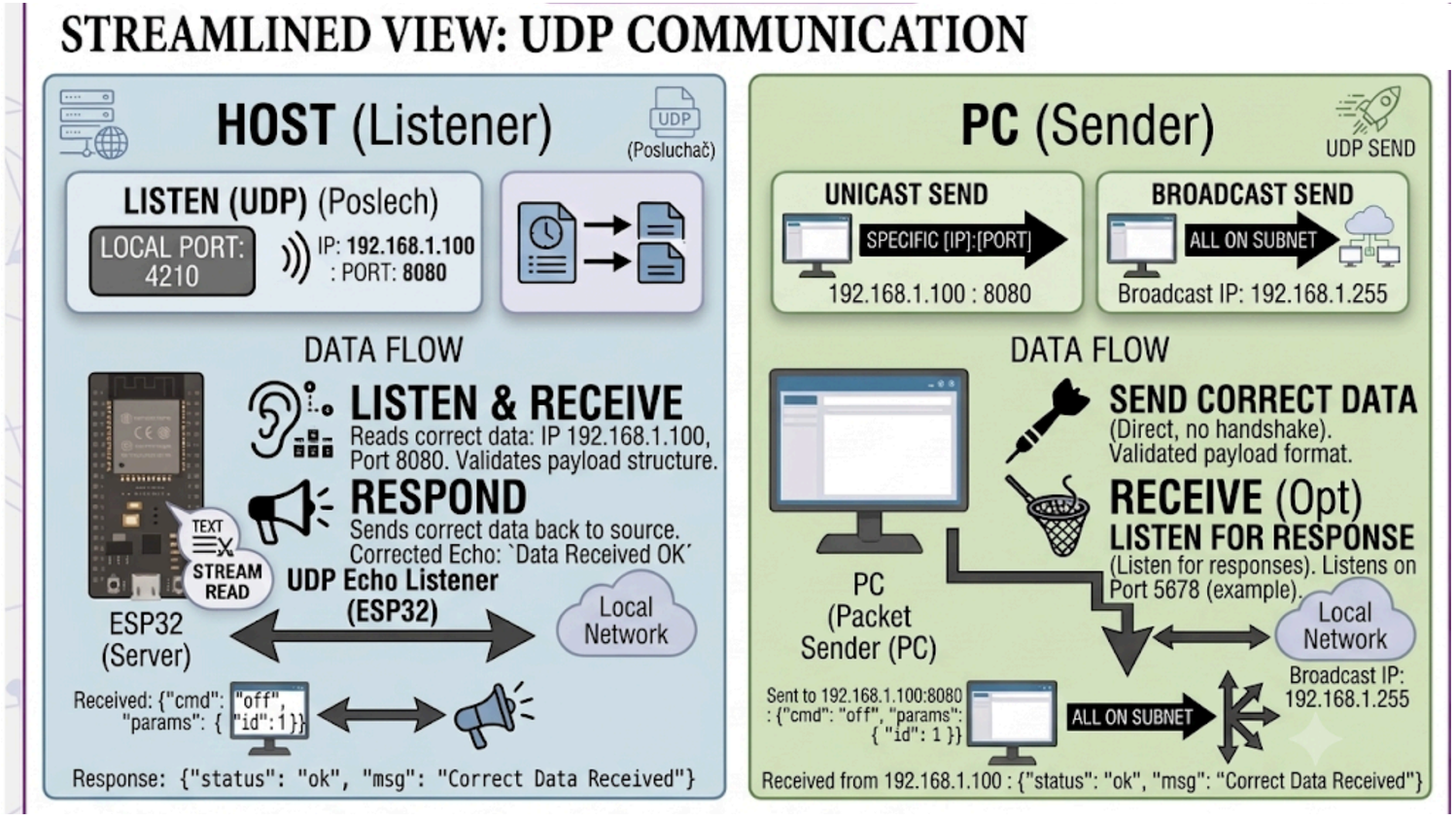
Step 5: Verify Communication & Echo Response in Packet Sender and Serial Monitor.

UDP Socket Eco Flow



UDP Communication Roles Comparison

A UDP Server/Receiver listens for unacknowledged incoming datagrams on a bound local port, while a UDP Client/Sender packages raw data into independent packets and sends them directly to a destination IP address and port.



A UDP transmission relies on exact socket endpoints defined by a target IP address and Port number (e.g., 192.168.1.50:4210) rather than HTTP URL paths. Unlike streaming protocols, UDP preserves distinct packet boundaries, so client transmissions should format payloads clearly or include explicit string delimiters (such as \n or a null terminator \0) to allow the receiver to cleanly parse the incoming buffer.

That's UDP Communication!

The access the implementation guide
Scan the QR / click on the link in the description

Thank You